

Constantin Seibold, Ph.D.

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Professional Experience 🏢

April 2026 - present	Lecturer , Department of Computational Linguistics, Heidelberg University
August 2025 - present	Research Group Leader , Diagnostic and interventional radiology, University Clinic Heidelberg
May 2023 - Nov. 2024	Junior Research Group Leader , Institute for Artificial Intelligence in Medicine, Clinic for Nuclear Medicine, University Medicine Essen
July 2019 - Dec. 2022	Research Assistant , Computer Vision for Human-Computer Interaction Lab, Karlsruhe Institute of Technology
March 2019 - July 2019	Student Research Assistant , Computer Vision for Human-Computer Interaction Lab, Karlsruhe Institute of Technology

Education 🎓

July 2019 - April 2023	PhD Computer Science , Karlsruhe Institute of Technology, Helmholtz Information & Data Science School for Health (HIDSS4Health) Thesis Title: Towards the automatic generation of medical reports in low supervision scenarios Grade: Summa Cum Laude , 🏆 KIT Doctoral Award Advisor: Prof. Dr.-Ing. Rainer Stiefelhagen
Oct. 2016 - March 2019	M. Sc. Computer Science , Karlsruhe Institute of Technology Thesis Title: Generative Models for Make-Up Style Transfer and Removal Master Thesis Advisor: Dr.-Ing. Muhammed Saquib Sarfraz, Prof. Dr.-Ing. Rainer Stiefelhagen
Oct. 2013 - Oct. 2016	B.Sc. Computer Science , University of Stuttgart Thesis Title: Improving author co-citation analysis in scientific literature by using citation function classification Bachelor Thesis Advisor: Dr.-Ing. Roman Klinger, Prof. Dr.-Ing. Sebastian Pado

Funded Research Projects 🏗️

2026	Medical Data Scientist Program , POLARIS: Platform for Optimizing Language-based Assistive Radiology Information Systems, Role: PI, Volume: 264.000 (100% E14 Position, 3 years)
2025	Seedfunding des Camilla-und-Georg-Jellinek-Zentrum für Ethik , Umgang mit unsicherem Wissen bei der Forschung zu krankheitsrelevanten Prädiktoren in der KI-gestützten medizinischen Bildgebung im Rahmen des Lungenkrebscreenings, Role: Support in Grant Application Process, Volume: 30.000€
2024	EU HORIZON JU - Innovative Health Initiative - HORIZON-JU-IHI-2023-05-02 , Thera4Care - THERANOSTICS ECOSYSTEM FOR PERSONALISED CARE, Role: PI, Sub-Project Volume: 260.000€ [Link]
2022	HIDSS4Health , Using Anatomical Knowledge to Improve Medical Image Analysis, Role: Support in Grant Application Process, Volume: 130.000€ (50% PhD position, 3 years) [Link]
2022	HIDSS4Health , Interactive Annotation of Volumetric Imaging Data Incorporating Report Information, Role: Support in Grant Application Process, Volume: 130.000€ (50% PhD position, 3 years) [Link]
2019	HIDSS4Health , Generating Medical Imaging Reports from 3D Radiological CT Scans using Image Captioning Techniques, Role: Team-member, project implementation[Link]

Patents

2024 **Aktenzeichen: 10 2023 112 553.2**, Computerimplementiertes Verfahren zum Trainieren eines maschinellen Lernmodells für die semantische Bildsegmentierung [DPMA]

Awards

2024 **2nd Place**, Automated Lesion Segmentation in Whole-Body PET/CT Challenge - autoPET III [paper]

2024 **1st Place (Spanish, Chinese), 3rd Place English**, MEDIQA-M3G Challenge at NAACL-ClinicalNLP [paper]

2023 **Winning Team of the UME - Innovation Contest**, University Medicine Essen

2023 **KIT Doctoral Award**, Karlsruhe Institute of Technology [link]

2022 **Student Travel Award**, Medical Image Computing and Computer Assisted Intervention Society [link]

2022 **Teaching Award - Best Practical Course**, Karlsruhe Institute of Technology, CS Faculty [link]

2019 **Best Industry Paper Award**, British Machine Vision Conference [paper]

Teaching Experience

THESES

Since 2024 **Supervision of Ph.D. Theses**, UME; Students: H. Kalisch

2023-2024 **Co-Supervision of Ph.D. Theses**, UME; Students: L. Heine, A. Brehmer, F. Jonske

2020-2024 **Supervision of Master Theses**, KIT, UME; Students: P. Nguyen, W. Di, P. Albrecht, R. Chlebecec

2020, 2024 **Supervision of Bachelor Theses**, KIT, UME; Students: C. Goos, J. Nasimzada, S. Mahler

COURSES

2026 **Projektseminar: Identifying Structure in Data**, University Heidelberg; Lecturer [SS26]

2022 **Deep Learning for Computer Vision I: Basics**, KIT; Teaching Assistant [SS22]

2021-2023 **Deep Learning for Computer Vision II: Advanced Topics**, KIT; Teaching Assistant, Lecturer [WS 22, WS 23]

2021-2022 **Practical Course - Computer Vision for Human Computer Interaction**, KIT; Lecturer/Course Organizer [ **Best Practical Course of SS21**] [WS21, SS21, WS22], consistently LQI 100 ("Lehrqualitätsindex", 100 is the best possible value at KIT).

2020-2021 **Deep Learning for Computer Vision**, KIT; Teaching Assistant [SS20, SS21]

Community Engagement & Outreach

2026-2027 **Marsilius Kolleg**, Young Marsilius Fellow, 5000 Euro Stipend

2026-current **ELLIS - European Laboratory for Learning and Intelligent Systems**, Member

2026-current **Camilla-und-Georg-Jellinek-Zentrum für Ethik**, Member

2025 **CVPR Tutorial - Identifying Structure in Data**, Organizer [link]

2023-2024 **Initiating and organizing the Computer-Vision Reading Group at the IKIM**, Organizer, Presenter

2021-2022 **Initiating and organizing the Computer-Vision Reading Group at CV:HCI**, Organizer, Presenter

2022 **MICCAI Workshop - Medical Applications with Disentanglements**, Program Committee

Invited Talks

- 2026 **Seoul, Korea**, KAIST ICML
Panel Discussion on Language Model Reliability & Safety
- 2025 **#algorithm-meeting @ DeepHealth**, Online Dimensionality Reduction and Clustering Tutorial
- 2025 **AI, ML and Computer Vision Meetup**, Online (hosted by voxel51)
Clustering in Computer Vision: From Theory to Applications
- 2024 **ENBIS Spring Meeting**, Dortmund, Germany
Interpretable AI in Medicine: Generating Radiological Reports with Panoptic Scene Graphs
- 2024 **Intelligent Sensing and Perception Group**, Stuttgart, Germany
The Current state of vision-language models in Radiology

Research Datasets & Software

PUBLIC DATASETS

- 2022–2023 **PAXRay / PAXRay++**, First author. 10,021 thoracic CTs, 157 anatomical labels projected to radiograph geometry; enables dense anatomy segmentation without manual annotation. BMVC 2022; extended 2023. [data]
- 2023 **DAP Atlas**, Co-first author. Anatomical masks for 533 CT scans covering 142 structures, built on AutoPET. ICIIP 2024. [paper]
- 2025 **MedShapeNet**, Co-author. Over 100,000 medical 3D shapes – bones, organs, vessels, surgical instruments – for translating computer-graphics methods to clinical use. [data]
- 2022 **Synthetic Structured Visual Content (SSVC)**, Co-author. 12,000 images of graphs and flow charts with bounding boxes and graph ground truth; supports machine reading of diagrams and accessibility for visually impaired readers. ICPR 2022.
- 2021 **Sims4Action**, Co-author. Ten hours of simulated activities of daily living across eight characters, camera angles and environments. IROS 2021.
- 2019 **Facial Cosmetic Content (FCC)**, Co-author. 18,425 facial images across strong, mild and no makeup, in 256×256 and 512×512 subsets. BMVC 2019 (Best Industry Paper).

OPEN-SOURCE SOFTWARE

- since 2023 **CXAS – Chest X-Ray Anatomy Segmentation**, Maintainer. PyPI package for fine-grained anatomy segmentation in chest radiographs. 28k downloads, [code]
- 2024 **Spacewalker**, Senior author. Interactive exploration and annotation of unstructured data. MLVis@EuroVis 2024. [code]
- since 2024 **2DAnatomyDatasets**, Maintainer. Curated collection of 2D anatomy-segmentation datasets. [code]
- since 2020 **SGL – Self-Guiding Loss**, First author. Reference implementation, ACCV 2020. [code]

Further Education

- 2023 **Workshop KHYS Basics of Leadership in Science 2023**, engaged with essential concepts of leadership in a scientific field.
- 2023 **Workshop KHYS Time-& Self-Management 2023**, presented approaches and tools for improved organization and self-reflexion.
- 2022 **IBM Neuro-Symbolic AI Summer School**, displayed possible combinations of knowledge-driven, symbolic AI with more traditional data-driven machine learning approaches. Distinguished speakers shared an overview of neuro-symbolic AI and how these method are applied in current applications.
- 2021 **Workshop Nawik Visualizing Science**, provided insights on how to properly visualize research results to convey the consequent insights.
- 2021 **Workshop Nawik Communicating Science**, provided insights on how to properly communicate your research results to both experts and layman.